

E-commerce Sales & Customer Insights

Jan 2023 – Sep 2025 · 18,246 orders · 6,000 customers · 98 products · 14 cities

About the data: no real dataset was provided, so this analysis uses a synthetic dataset generated by `src/generate_data.py` (fixed seed, deliberately messy). Figures illustrate the method, not a real business.

Net revenue

₹5.42 Cr

Orders

16,075

Avg order value

₹3,370

Repeat-customer
rate

47.5%

Gross margin

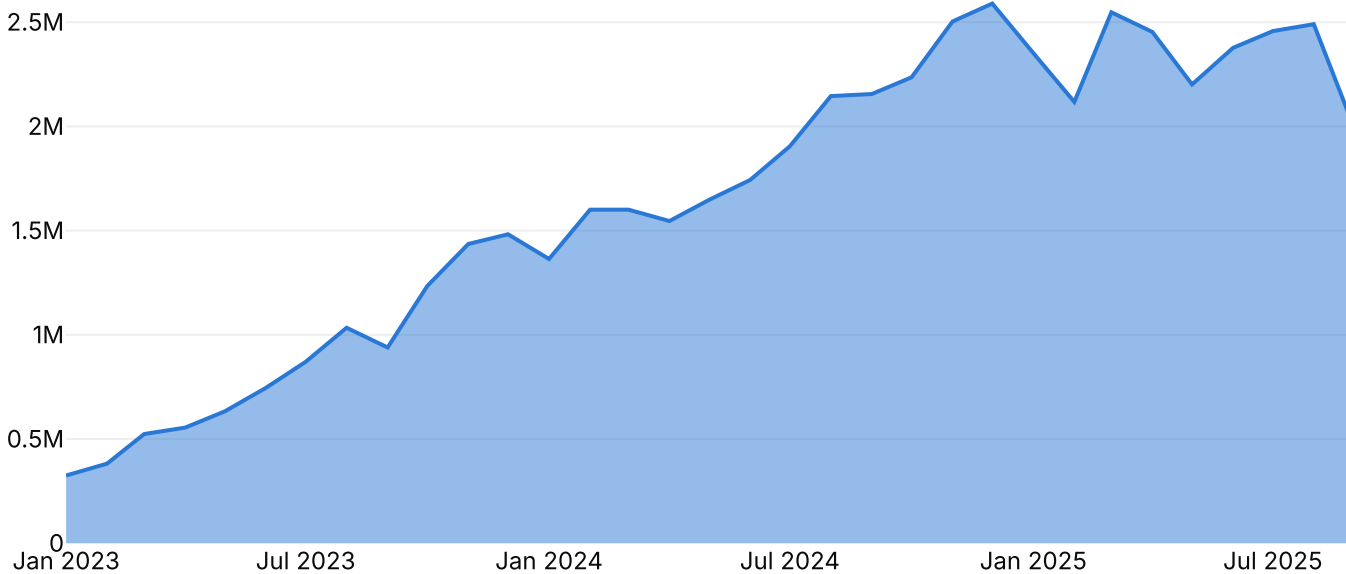
22.5%

Return rate

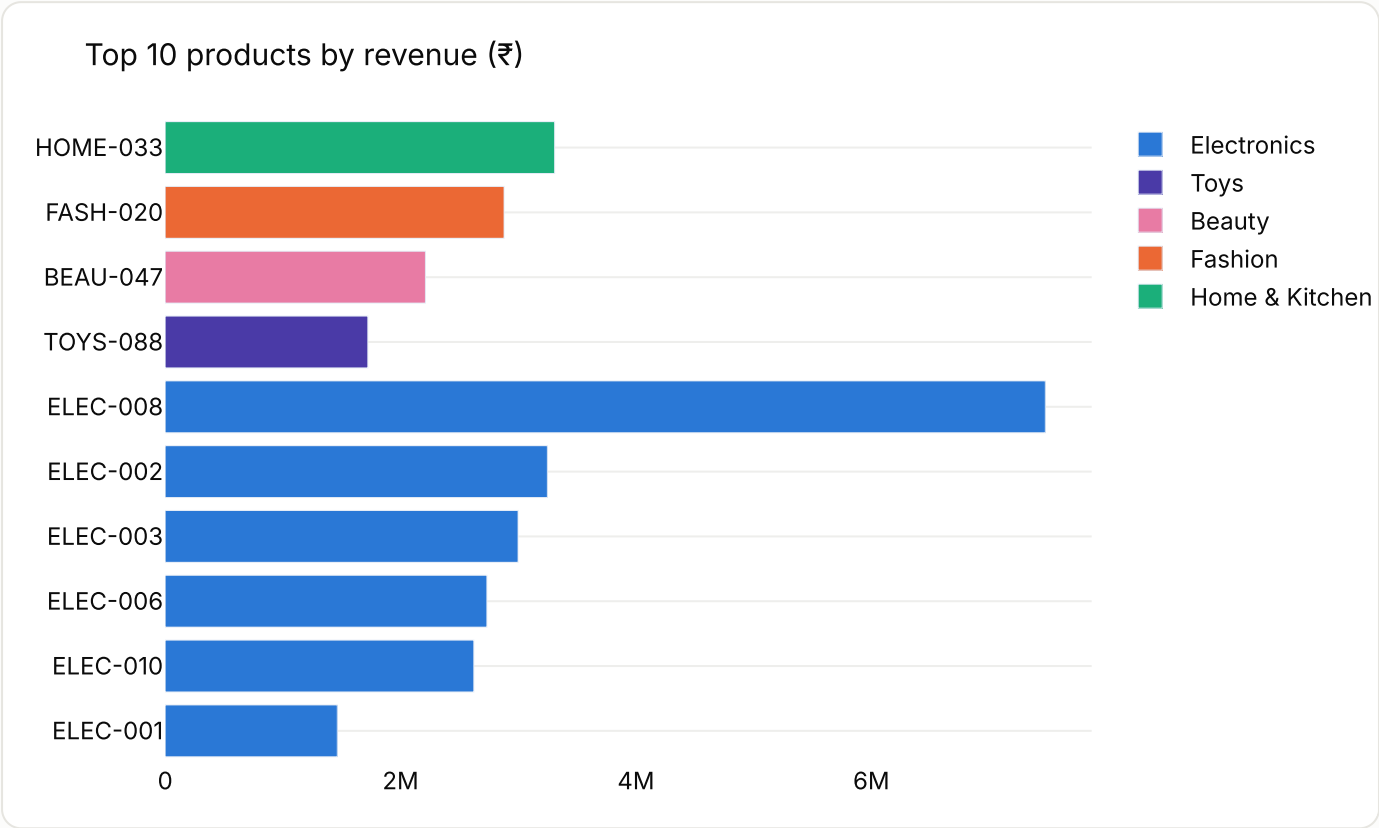
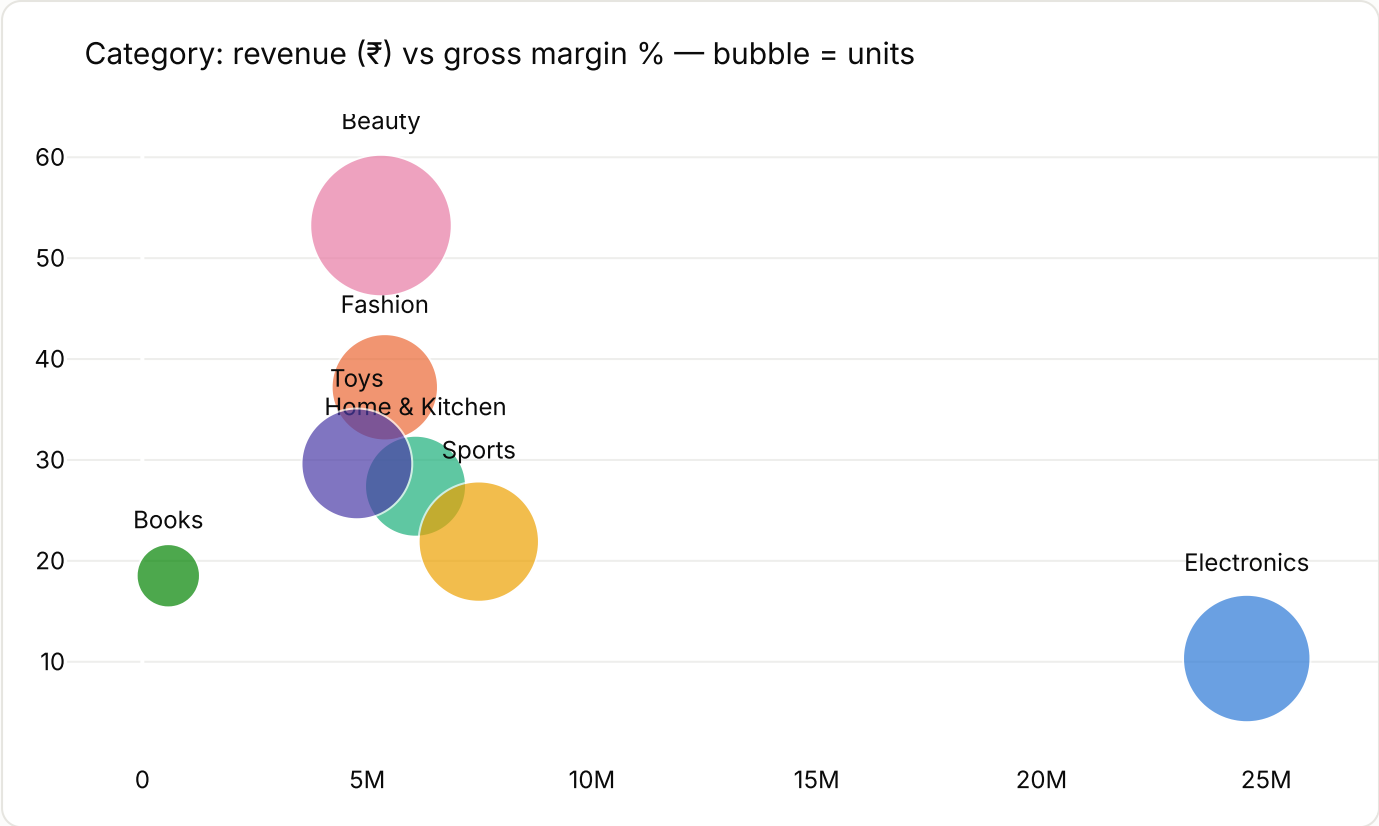
8.0%

1. Revenue trend

Monthly net revenue (₹)

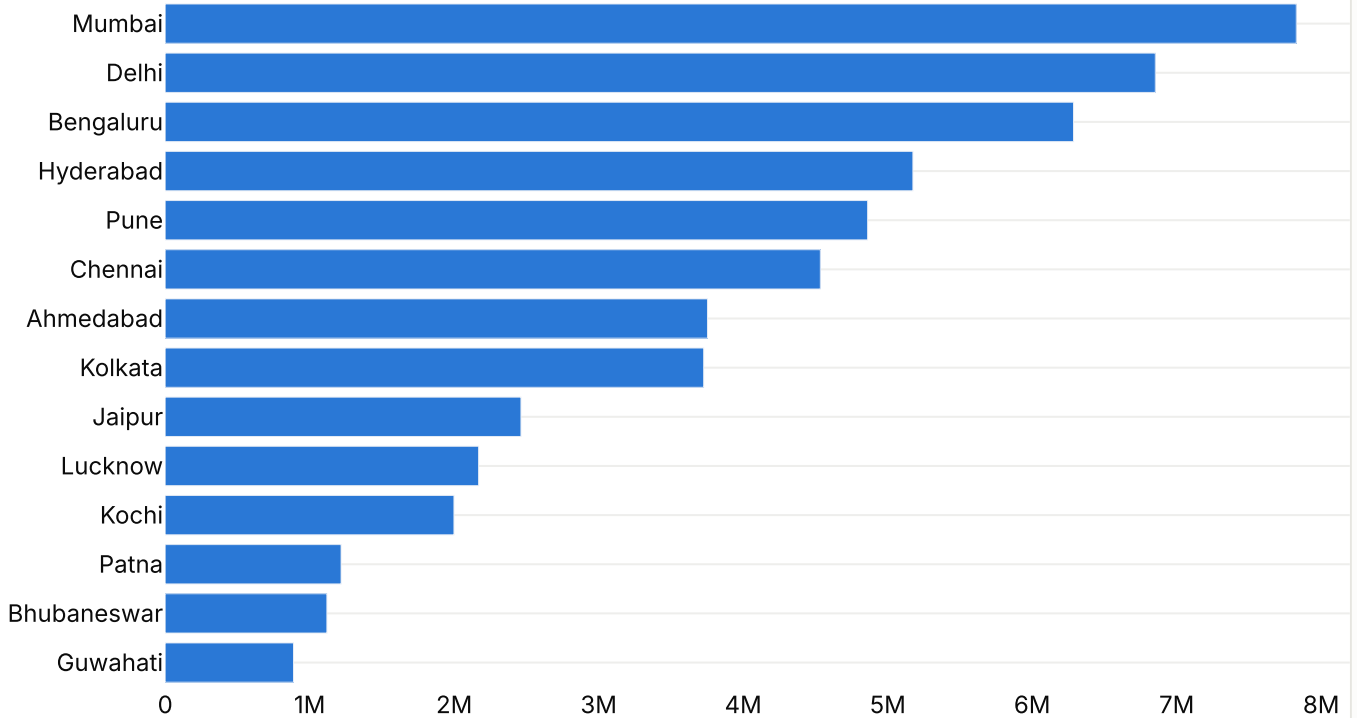


2. Products & categories



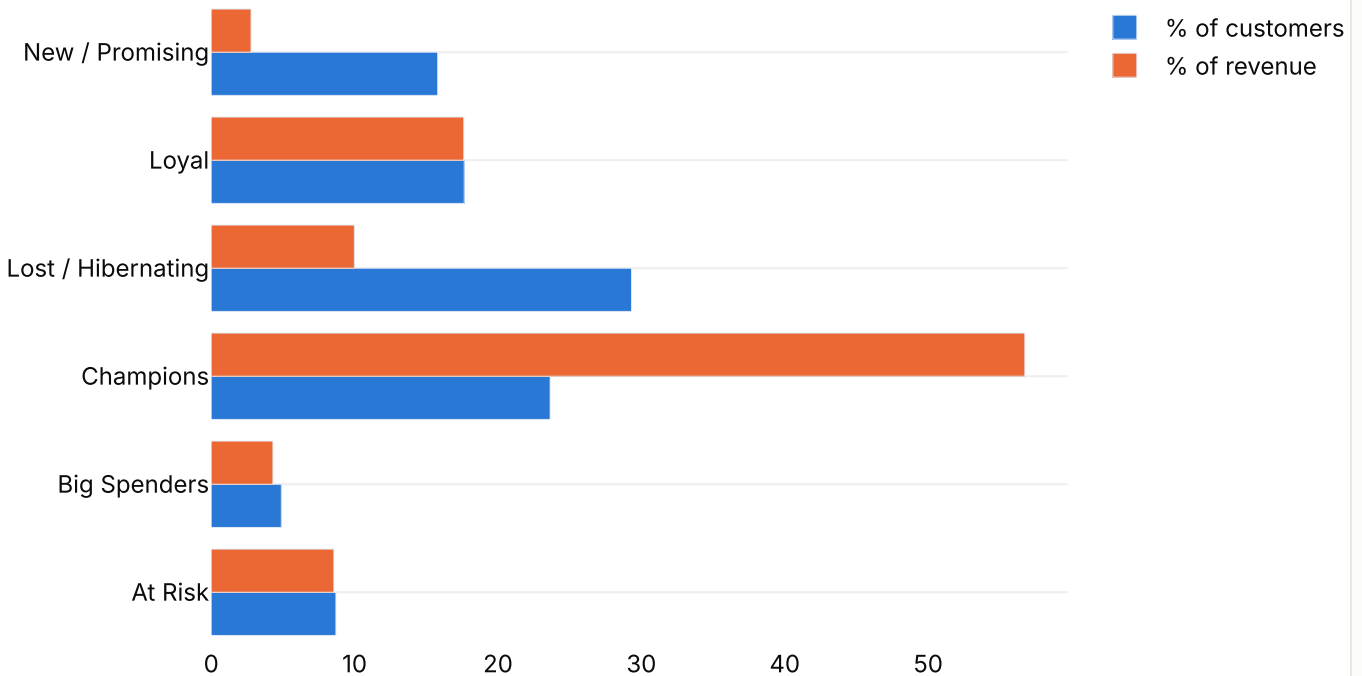
3. Geography

Revenue by city (₹)

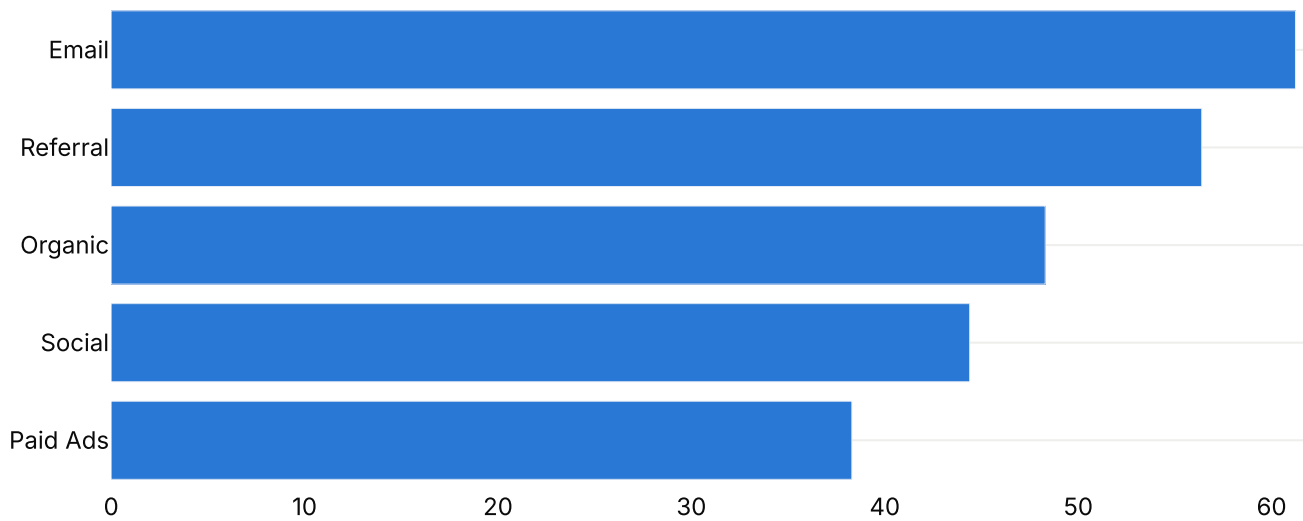


4. Customer segments

RFM segments: share of customers vs share of revenue (%)

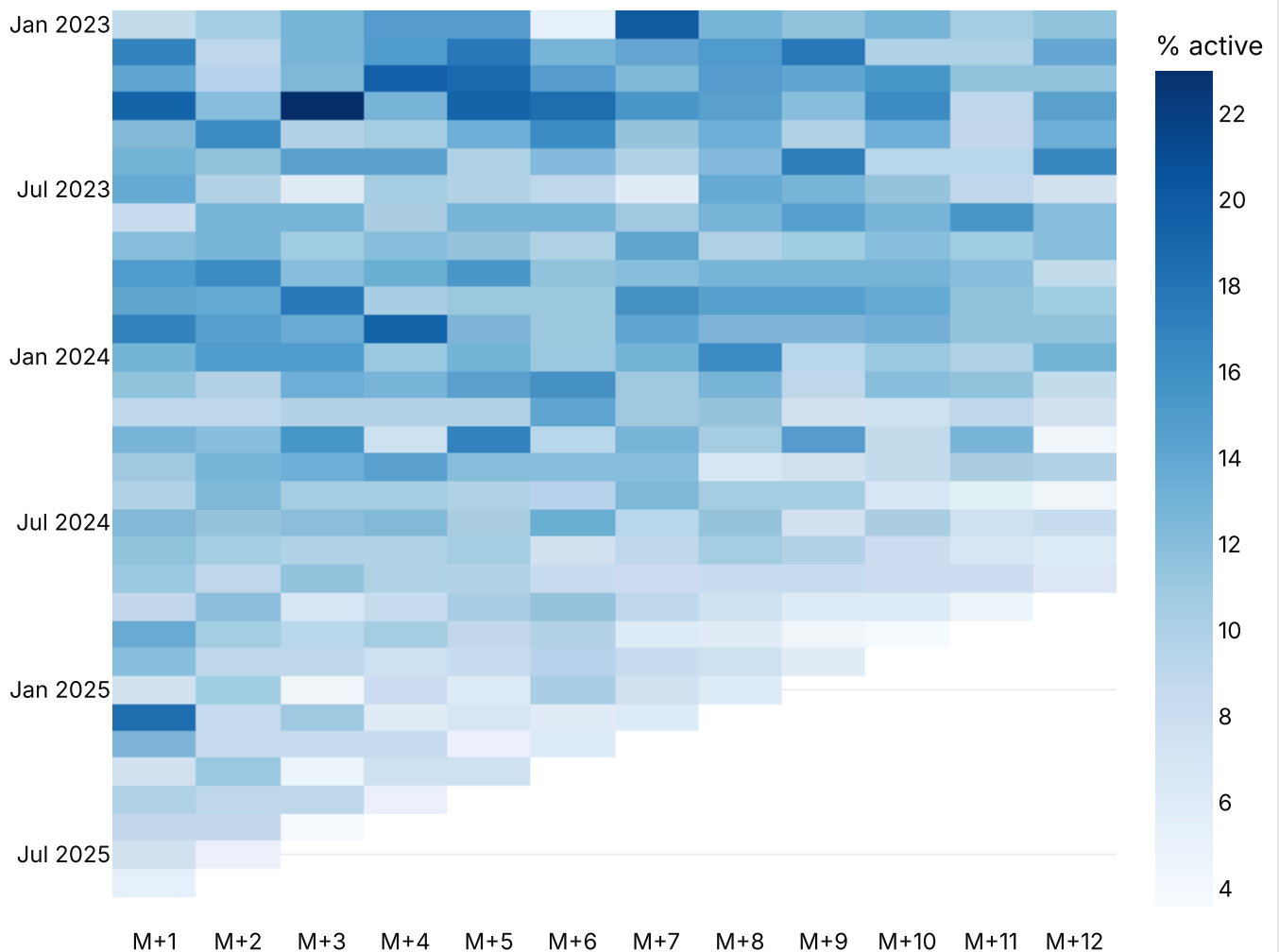


Repeat-customer rate by acquisition channel (%)



5. Cohort retention

Cohort retention — % of signup-month cohort ordering N months later



6. Recommendations

1. **Rebalance towards margin.** Electronics is the biggest category (₹2.46 Cr, 45% of sales) but earns only 10.3% margin vs 53.2% for Beauty. Cross-sell Beauty to Electronics buyers and review Electronics discounting.
2. **Fix or prune Books.** It brings only ₹5.78 L (1.1% of revenue) at 19% margin — bundle it or cut slow SKUs.
3. **Buy quality, not volume, in acquisition.** Email customers repeat at 61% vs 38% for Paid Ads. Shift budget towards Email and tighten Paid Ads targeting.
4. **Protect Champions.** 1,329 customers (24% of base) generate 57% of revenue — add loyalty tiers and early access.
5. **Win back At-Risk buyers.** 489 formerly frequent customers have gone quiet; a time-limited win-back email is the cheapest revenue available.
6. **Grow beyond the core.** South and West deliver 65% of revenue. East and North-East are under-penetrated — test local campaigns and free-shipping thresholds.
7. **Attack returns where refunds concentrate.** 8.0% of orders are returned (₹48.25 L refunded); Electronics is 43% of refunded value.

Method

Order, customer, product and location tables were merged into one order-line table. Cleaning steps are logged below. Revenue counts **Delivered** orders only; **Returned** orders are treated as refunds and **Cancelled** orders never complete. Net revenue = quantity × price × (1 – discount); margin uses unit cost. Repeat-customer rate = customers with 2+ delivered orders. Customers are segmented with RFM quartiles (recency, frequency, monetary).

step	rows_affected	detail
Drop duplicate orders	274	same order_id repeated
Drop duplicate order lines	284	exact repeated rows
Standardise product categories	5	12 spellings → 7 categories
Standardise status / payment labels	0	case + whitespace fixed

step	rows_affected	detail
Drop non-positive quantities	12	data-entry errors
Drop future-dated orders	8	after 2025-09-30
Impute missing values	0	age median (240), gender/payment 'Unknown', location 'UNK' (120), discount 0 (294)
Drop lines without a valid order	16	orphans after order cleaning
Final analysis table	0	18,246 orders, 6,000 customers